Counting Sasquatch: A Bayesian Analysis of Reported Bigfoot Sightings

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1 Introduction

Bigfoot sightings are a serious problem in North America: widely reported, never verified, and still somehow abundant enough to produce a usable dataset. This study does not attempt to determine whether Sasquatch is real, which would require stronger evidence than a count model and a CSV file. Instead, it studies the phenomenon that is unquestionably real: people report seeing Bigfoot, and those reports occur at specific times that can be analyzed statistically. **The research question is whether reported Bigfoot sightings show systematic temporal structure rather than appearing as random noise.**

To study this, we use Bayesian count regression, beginning with a **Poisson model** and then extending to a **Negative Binomial model**. These models are designed for nonnegative integer outcomes and relate the expected count to predictors through a log link, allowing seasonal patterns and long-run time trends to be estimated while keeping fitted counts positive. In this application, the predictors are month-of-year effects and time trends, so the model can assess whether sightings cluster in particular seasons and whether reporting rises or falls over time.

The Poisson model is a natural starting point, but it assumes that the conditional variance equals the conditional mean. That assumption is often too restrictive for data that arrive in bursts. Bigfoot reports are a plausible example: sightings may vary because of outdoor activity, hunting seasons, local folklore, media attention, or an unusually productive month for mistaking bears for cryptids. To account for this extra variability, the study also considers a Negative Binomial model, which adds an overdispersion parameter and is more flexible when the counts fluctuate more than a Poisson model can reasonably accommodate.

The analysis is conducted in a Bayesian framework, which treats unknown parameters as random variables and combines prior distributions with the observed

data to obtain posterior distributions. This provides direct uncertainty quantification for seasonal effects, time trends, and overdispersion, while also making prior sensitivity analysis straightforward. The models apply only to the *reporting process*, not to the biological existence of Bigfoot, as illustrated in Figure 1. They cannot distinguish genuine encounters from hoaxes, folklore, misidentification, or ordinary human enthusiasm for mysterious forest creatures. What they can do is answer a more focused question: whether the Bigfoot sighting phenomenon is patterned, predictable, and statistically interesting.

Figure 1: Bigfoot depiction from the 1967 Patterson–Gimlin film

2 Dataset Description

The dataset used in this project consists of reported Bigfoot sightings from the Bigfoot Field Researchers Organization (BFRO), stored in a clean CSV file containing one row per report. The data were obtained from the Kaggle repository "Unlocking Mysteries of Bigfoot Through Sightings".

The original file includes the variables `number` (report ID), `title` (short description of the report), `classification` (report credibility class), `timestamp` (date of the reported sighting), and geographic coordinates given by `latitude` and `longitude`. Since the goal of this study is to understand temporal patterns in reported sightings rather than spatial variation, the main variable of interest is the report date.

To focus on the more credible public reports, the analysis restricts attention to `Class A` and `Class B` sightings and excludes the small number of `Class C` reports. Implausible dates were also removed, leaving a final sample covering January 1950 through November 2021. The sighting-level data were then aggregated into monthly counts, producing a time series in which each observation represents the total number of reported sightings in a given month, as seen in Figure 2 and Figure 3.

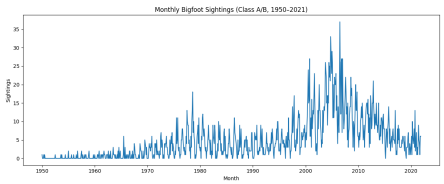

Figure 2: Time series plot of Bigfoot sightings

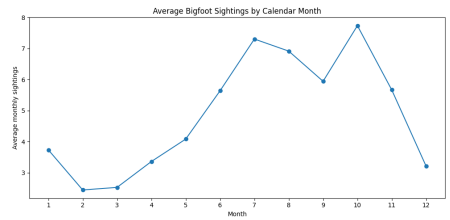

Figure 3: Average Bigfoot sightings by month

The resulting modeling dataset contains the variables `year_month`, `sightings`, `month`, `year`, `time_index`, and standardized time-trend variables. The response variable, `sightings`, is the number of Bigfoot reports in a month. The variable `month` captures recurring seasonal effects, while `time_index` and its standardized transformations capture long-run temporal trends.

The research question as stated in Section 1 is whether reported Bigfoot sightings exhibit systematic seasonality and long-run temporal structure, and whether these counts are better modeled by a Poisson or Negative Binomial process. To answer this, the monthly counts are analyzed using Bayesian count regression with month-of-year effects and time trends, allowing the study to assess clustering by season, changes over time, and overdispersion beyond what a Poisson model can capture.

3 Explanation of Prior Choices

3.1 Prior 1: Weakly Informative

The main analysis uses *weakly informative priors*, chosen to regularize estimation without imposing overly strong beliefs about the size or direction of the effects. Since the models use a log link, the intercept and regression coefficients are defined on the log expected-count scale. For the primary specification, the intercept was assigned

$$\alpha \sim \mathcal{N}(0, 3),$$

while the time-trend coefficients were assigned

$$\beta_{\text{time}}, \beta_{\text{time}^2} \sim \mathcal{N}(0, 1).$$

Month effects were constructed from raw parameters with

$$\gamma_m^{\text{raw}} \sim \mathcal{N}(0, 1),$$

and then centered to satisfy a sum-to-zero constraint so that each month effect is interpreted relative to the average month. These priors are broad enough

to allow meaningful seasonal and temporal effects, but still discourage implausibly extreme values. For the Negative Binomial models, the overdispersion parameter was assigned

$$\alpha_{NB} \sim \text{Exponential}(1),$$

which enforces positivity while remaining flexible about the amount of extra-Poisson variation.

3.2 Prior 2: Higher Regularization

A second prior specification was used as a sensitivity analysis to assess whether the conclusions depended heavily on prior scale. This more regularizing specification used

$$\alpha \sim \mathcal{N}(0, 2), \qquad \beta_{\text{time}}, \beta_{\text{time}^2} \sim \mathcal{N}(0, 0.5), \qquad \gamma_m^{\text{raw}} \sim \mathcal{N}(0, 0.5),$$

with

$$\alpha_{NB} \sim \text{Exponential}(0.5).$$

By shrinking coefficients more strongly toward zero, these priors provide a useful check on whether the posterior patterns are driven by the observed data or by weak prior assumptions. This is especially relevant in a topic like Bigfoot, where one would prefer not to be accused of smuggling Sasquatch into the posterior through an overly enthusiastic prior.

The two prior specifications produced very similar posterior summaries across all three models. In the Poisson model, the posterior means for the intercept and time effect were essentially unchanged under the two prior sets. The same was true in the linear Negative Binomial model, where the intercept, linear time coefficient, and overdispersion parameter were nearly identical across priors. The quadratic Negative Binomial model showed the same stability: the linear trend remained positive, the quadratic term remained negative, and the estimated overdispersion parameter changed very little, as seen below in Table 1

Table 1: Prior sensitivity comparison for key posterior means

Model	Parameter	Prior 1	Prior 2
Poisson	α	1.256	1.257
Poisson	β_{time}	0.732	0.732
NB linear	α	1.142	1.146
NB linear	β_{time}	0.995	0.987
NB linear	α_{NB}	2.207	2.230
NB quadratic	α	1.565	1.567
NB quadratic	β_{time}	1.205	1.195
NB quadratic	β_{time^2}	-0.611	-0.607
NB quadratic	α_{NB}	3.382	3.446

These comparisons suggest that the substantive conclusions of this study are not sensitive to moderate changes in prior scale. For that reason, the weakly informative prior specification is used as the main analysis, while the more regularizing prior set is treated as a robustness check.

4 Analysis Results

A preliminary feature of the data is clear *overdispersion*. After restricting the sample to Class A and Class B reports from 1950 through 2021 and aggregating to monthly counts, the mean number of sightings per month was approximately 4.88, while the variance was approximately 32.89. The variance-to-mean ratio was therefore about 6.74, far larger than 1, indicating that a Poisson model is likely too restrictive. Substantively, this is not surprising: reported Bigfoot sightings probably do not arrive as described in textbook Poisson processes. Instead, reports may cluster because of media attention, outdoor activity, local folklore, or bursts of enthusiasm about mysterious forest creatures. This provided the main motivation for comparing Poisson and Negative Binomial models.

Three Bayesian count models were estimated in PyMC: a Poisson model with month effects and a linear time trend, a Negative Binomial model with the same predictors, and a Negative Binomial model with month effects plus both linear and quadratic time terms. The three models can be written as follows. Let Y_t denote the number of reported Bigfoot sightings in month t, let $\gamma_{m(t)}$ denote the effect of the calendar month associated with observation t, and let x_t denote the standardized time index.

$$Y_t \sim \text{Poisson}(\mu_t), \tag{1}$$

$$\log \mu_t = \alpha + \beta_{\text{time}} x_t + \gamma_{m(t)}. \tag{2}$$

This is the baseline Poisson specification, in which the expected monthly count depends on a linear time trend and month-of-year effects.

$$Y_t \sim \text{NegBin}(\mu_t, \alpha_{NB}), \tag{3}$$

$$\log \mu_t = \alpha + \beta_{\text{time}} x_t + \gamma_{m(t)}. \tag{4}$$

This Negative Binomial model keeps the same mean structure as the Poisson model but introduces an overdispersion parameter α_{NB} to allow the variance to exceed the mean.

$$Y_t \sim \text{NegBin}(\mu_t, \alpha_{NB}), \tag{5}$$

$$\log \mu_t = \alpha + \beta_{\text{time}} x_t + \beta_{\text{time}^2} x_t^2 + \gamma_{m(t)}. \tag{6}$$

This final specification adds a quadratic time term, allowing the long-run trend in reported sightings to rise and then fall rather than remain purely linear.

In all three models, the month effects were centered to satisfy a sum-to-zero constraint, so that each γ_m is interpreted relative to the average month.

All models were fit using 4 chains, 2000 warmup iterations, and 2000 posterior draws per chain, with `target_accept = 0.90`. All showed satisfactory MCMC performance. Trace plots had solid mixing across chains, there were no divergent transitions, all $\hat{R}$ values were essentially 1.00, and effective sample sizes were large for all key parameters. Since these diagnostics indicated stable posterior exploration, no additional reparameterization or tuning beyond the chosen target acceptance rate was necessary.

Model comparison strongly favored the Negative Binomial specification with a quadratic time trend. Table 2 reports approximate leave-one-out cross-validation results. The quadratic Negative Binomial model achieved the highest expected log predictive density and received essentially all of the model weight, while the linear Negative Binomial model performed substantially worse and the Poisson model performed much worse still. This confirms that the sighting data are not only overdispersed, but also exhibit a nonlinear long-run pattern that is not captured well by a purely linear specification.

Table 2: Model comparison using approximate leave-one-out cross-validation

Model	Rank	$\widehat{\text{elpd}}_{\text{loo}}$	$\Delta\widehat{\text{elpd}}_{\text{loo}}$	Weight
Negative Binomial + quadratic trend	0	-1857.65	0.00	0.991
Negative Binomial + linear trend	1	-1961.47	103.82	0.000
Poisson + linear trend	2	-2473.79	616.15	0.009

The main posterior results from the preferred model are summarized in Table 3. The linear time coefficient was positive, while the quadratic coefficient was negative, implying a hump-shaped long-run trend rather than a monotone increase. The posterior probability that the quadratic coefficient was negative was 1.0, providing essentially complete posterior support for concavity in the time trend. Converting the quadratic coefficients into an implied turning point yielded a posterior median peak around June 2006, with a posterior interval running approximately from May 2004 to January 2009. Thus, the fitted model suggests that reported sightings rose for several decades, peaked in the mid-2000s, and then declined afterward.

Table 3: Key posterior summaries for the preferred Negative Binomial model with quadratic trend

Parameter	Posterior Mean	94% HDI Lower	94% HDI Upper
Intercept α	1.565	1.489	1.640
Linear time effect β_{time}	1.205	1.127	1.282
Quadratic time effect β_{time^2}	-0.611	-0.686	-0.527
Overdispersion α_{NB}	3.382	2.814	3.969

The estimated month effects showed strong and highly interpretable seasonality. Late summer and fall months were consistently above average, while late winter and early spring months were below average. In particular, July and October had some of the largest positive effects, whereas February and March had some of the most negative effects. These seasonal patterns were stable across the Poisson, linear Negative Binomial, and quadratic Negative Binomial models, and they also remained nearly unchanged under the alternative prior specification. Table 4 reports the posterior summaries for the month effects in the preferred model.

Table 4: Posterior month effects for the preferred Negative Binomial model with quadratic trend

Month	Posterior Mean	94% HDI Lower	94% HDI Upper
January	-0.215	-0.387	-0.039
February	-0.680	-0.861	-0.483
March	-0.643	-0.826	-0.463
April	-0.302	-0.476	-0.127
May	-0.121	-0.294	0.044
June	0.237	0.074	0.406
July	0.571	0.411	0.726
August	0.497	0.343	0.660
September	0.276	0.113	0.429
October	0.553	0.396	0.709
November	0.205	0.048	0.366
December	-0.379	-0.557	-0.201

Since the month effects are on the log-count scale and are centered to sum to zero, positive values indicate above-average expected sightings and negative values indicate below-average sightings, holding the long-run trend fixed. For example, October's posterior mean effect of 0.553 implies an expected sighting rate of about $\exp(0.553) \approx 1.74$ times the average month, while February's effect of -0.680 implies a rate of about $\exp(-0.680) \approx 0.51$ times the average month. This reinforces the pattern that sightings are highest in late summer and fall and lowest in late winter.

Overall, reported Bigfoot sightings are not well described by a simple Poisson process. They are overdispersed, strongly seasonal, and characterized by a non-linear long-run trend. The best-fitting model indicates that reports were higher in late summer and fall, lower in late winter and early spring, and rose before declining, with a peak in the mid-2000s. These results do not establish the existence of Bigfoot, but they do show that the sighting phenomenon is statistically structured rather than temporally uniform. Even in cryptozoology, the counts appear to keep a schedule.

5 Summary of Findings

This groundbreaking study found that reported Bigfoot sightings do exhibit systematic temporal structure rather than random noise. Monthly sighting counts are overdispersed relative to a Poisson process, display clear seasonality, and follow a nonlinear long-run trend.

Among the models considered, the **Negative Binomial model with a quadratic time trend** provided the best fit. This suggests that both extra-Poisson variability and a nonlinear time pattern are important features of the sighting process. Figure 4 shows that the preferred model captures the broad rise and decline in reported sightings more realistically than either the Poisson model or the linear Negative Binomial model.

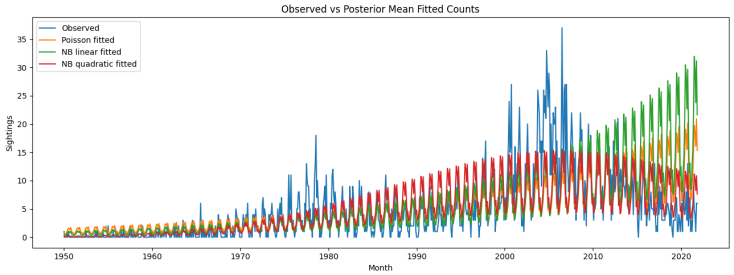

Figure 4: Observed monthly Bigfoot sightings and posterior mean fitted counts for the three Bayesian count models. The Negative Binomial model with quadratic time trend provides the best overall fit.

Posterior inference from the final model showed complete support for a negative quadratic coefficient, implying a hump-shaped long-run pattern. The implied peak occurred in the mid-2000s, with a posterior median near June 2006. The final model also revealed strong seasonality: late summer and fall months were above average, while late winter and early spring months were below average. July and October were among the strongest positive months, whereas February and March were among the most negative, as shown in Figure 5.

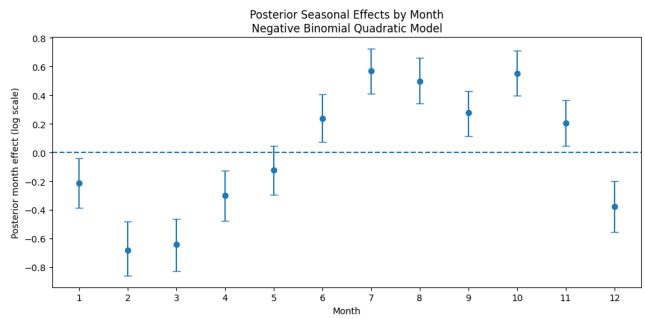

Figure 5: Posterior month effects for the preferred Negative Binomial model with quadratic time trend. Late summer and fall are above average, while late winter and early spring are below average.

Overall, these results imply that reported Bigfoot sightings are better understood as an overdispersed, seasonally patterned reporting process with a nonlinear long-run trajectory. This does not establish the biological existence of Bigfoot, but it does show that the sighting phenomenon itself is structured, predictable, and statistically nontrivial. In other words, Sasquatch may remain elusive, but the overdispersion is not.

6 Extension: Estimating the Peak of Bigfoot Sightings

The preferred Negative Binomial model with a quadratic time trend was used to estimate the *implied peak date* of reported Bigfoot sightings. Since the fitted trend is quadratic in standardized time, the turning point occurs at

$$x^* = -\frac{\beta_{\text{time}}}{2\beta_{\text{time}^2}},$$

which can be transformed back to the original time scale and then into a calendar month. Applying this transformation to posterior draws of β_{time} and β_{time^2} yields a posterior distribution for the date at which the fitted long-run sighting trend reached its maximum.

The results provide strong support for a concave time pattern. The posterior probability that the quadratic coefficient is negative was

$$P(\beta_{\text{time}^2} < 0 \mid \text{data}) = 1.0,$$

and the posterior median peak date was **June 2006**, with a posterior interval extending from approximately **May 2004** to **January 2009**. The posterior mass is concentrated heavily in the mid-2000s, as shown in Figure 6, indicating that the reporting process was most intense during that period.

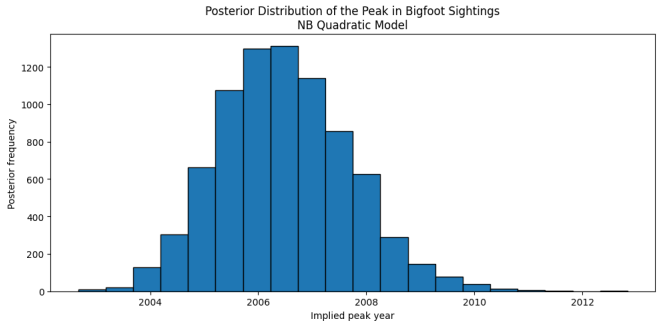

Figure 6: Posterior distribution of the implied peak year in reported Bigfoot sightings under the preferred Negative Binomial quadratic model.

This extension does not imply that Bigfoot populations literally surged and declined like a poorly monitored wildlife species. It implies only that the *reporting phenomenon* itself appears to have peaked around 2006. Even so, the result provides a useful Bayesian summary: the model does not merely indicate that sightings changed over time, but gives a probabilistic answer to when the era of peak Sasquatch reporting most likely occurred. If there was ever a golden age of Bigfoot paperwork, the posterior suggests it was around 2006.